

Kinetic Theory of Gases

→ Molecular nature of matter = The Intermolecular space is very less between the molecules of Solids and it is Very Large between the molecules of gas.

→ When molecule of gas collides with any molecules once and then with other neither speed nor the direction of molecules remain constant.

matter → molecules

Intermolecular distance

↓
Solid liquid gas

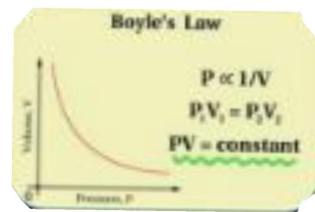
→ Intermolecular force =
Solid > gas

→ Equation of state of a perfect gas =

① Boyle's Law = At constant Temperature volume of gas is inversely proportional to pressure.

$$P \propto \frac{1}{V}$$

$$PV = \text{Constant}$$



(2) Charles's Law = If pressure is constant then Volume of gas directly proportional to temperature

$$V \propto T, \frac{V}{T} = \text{Constant}$$

$$\Rightarrow \frac{PV}{T} = \text{Constant}$$

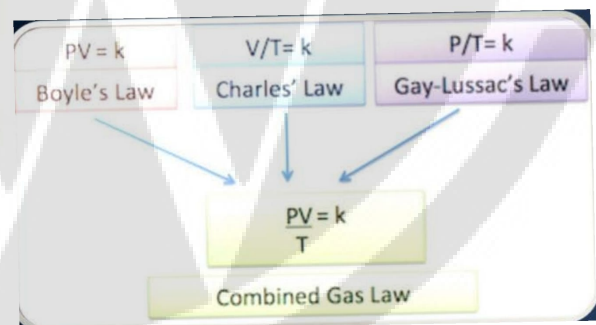
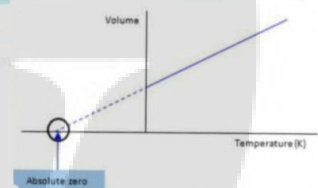
$$= \frac{PV}{T} = R$$

$$PV = RT$$

R = Universal gas Constant

R = 8.31 Joule/molK

Ideal gas equation



→ Ideal gas = Condition for Ideal gas

(i) molecules should be infinitely small and at very far distance

(ii) There is no mutual attraction b/w molecules

Total Internal energy = due to motion [K.E.]

Internal pot. energy = 0

→ How can we get Ideal gas?

→ At High Temperature and

→ At low pressure Real gas Behave as an Ideal gas

Temp ↑↑
gap b/w Real and Ideal gas ↓↓

→ Ideal gas equation =

$$PV = nRT$$

(n = no of moles)

no. of moles = $\frac{\text{no. of molecules}}{\text{Avogadro no}}$
(n)

$$n = \frac{N}{N_A}$$

$$PV = \frac{N}{N_A} RT$$

→ Boltzmann's Constant =

$$k_B = \frac{R}{N_A}$$

$$PV = N k_B T$$

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

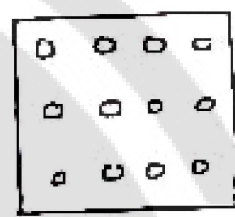
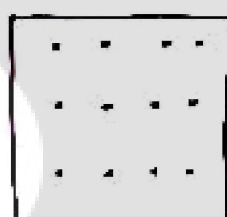
$$N_A = 6.028 \times 10^{23}$$

$$R = 8.31 \text{ Joule/moleK}$$

→ Avogadro Number =

Number of molecules in one mole of material.

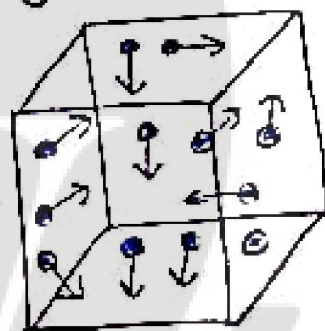
→ Hypothesis At Same pressure and temperature equal Volume of all gases contain equal number of molecules



→ Kinetic Theory of gases -

Assumptions =

① Every gas is made up of small particle called molecule



② molecule should be same size and mass

③ All the molecules should be at rest and very small.

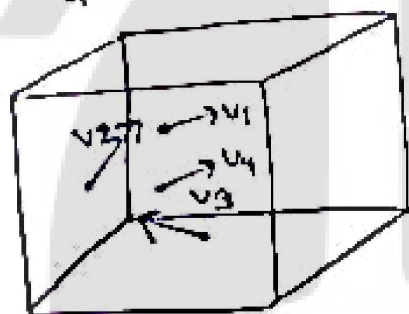
④ Collision of molecules should be perfectly elastic.

⑤ There should not be mutual attraction and repulsion b/w molecules

(6) The average kinetic energy of gas molecules depends only on temperature and is the same for all gases at the same temperature.

→ Root mean Square Velocity of molecules =

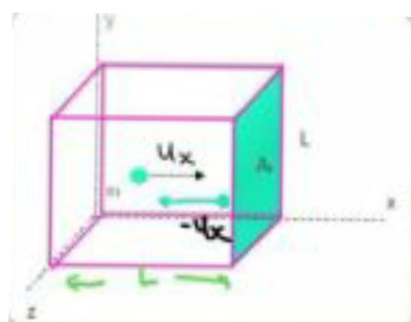
Let the velocities of n molecules are $v_1, v_2, \dots, v_n$



$$v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2}{n}}$$

→ The rms Speed (v_{rms}) is the square root of the mean of molecular velocities.

→ Expression for pressure Exerted by a gas =



→ Let a molecule is moving in x direction so momentum

$$p_{initial} = m u_x$$

After collide it bounce back with speed $-u_x$

$$p_{final} = -m u_x$$

change in momentum

$$\Delta p = p_f - p_i = -m u_x - m u_x$$

$$\Delta p = -2m u_x \text{ or } \Delta p = 2m u_x \text{ --- (1)}$$

If time taken is Δt in change in momentum Δp according to NLM 2nd Law

$$F = \frac{\Delta p}{\Delta t}$$

$$F = \frac{2m u_x}{t} \text{ --- (2)}$$

$$\text{time} = \frac{\text{distance}}{\text{velocity}}$$

$$\Delta t = \frac{2L}{u_x}$$

put value of Δt in equation (2)

$$F = \frac{2m u_x}{(2L/u_x)}$$

$$F = \frac{m u_x^2}{L} \text{ --- (3)}$$

pressure Exerted by this molecule $p = \frac{F}{A}$

$$p = \frac{m u_x^2}{L A}$$

$$p = \frac{m u_x^2}{L L^2}$$

$$\left[A = L^2 \right] \\ \left[\text{volume} = L^3 \right]$$

$$p = \frac{m u_x^2}{V} \text{ --- (4)}$$

molecules move in complete volume

$$\vec{u} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k}$$

→ magnitude

$$u^2 = u_x^2 + u_y^2 + u_z^2$$

$$u^2 = u_x^2 + u_y^2 + u_z^2 \quad [u_x = u_y = u_z]$$

$$u^2 = 3u_x^2$$

$$u_x^2 = \frac{u^2}{3} \quad \text{--- (5)}$$

→ pressure = $p = \frac{m u_{rms}^2}{3V}$

If density is ρ

$$\rho = \frac{m}{V}$$

$$p = \frac{1}{3} \rho u_{rms}^2$$

→ If there are n molecules
pressure $p = \frac{1}{3} \frac{mn}{V} u_{rms}^2$

$$[\text{Total mass } m = mn]$$

$$p = \frac{1}{3} \frac{M}{V} u_{rms}^2$$

$$p = \frac{1}{3} \rho u_{rms}^2$$

All molecules exert pressure on the wall of container.

→ Kinetic energy of one molecule =

$$KE = \frac{1}{2} m u^2$$

Kinetic energy of one molecule per unit volume

$$KE = \frac{1}{2} \frac{m u^2}{V}$$

Kinetic energy of n molecule

$$KE = \frac{1}{2} m n u_{rms}^2$$

Kinetic energy of n molecule per unit volume

$$KE = \frac{1}{2} \left(\frac{mn}{V} \right) u_{rms}^2$$

Kinetic energy of n molecules per unit volume

$$KE = \frac{1}{2} \left(\frac{mn}{V} \right) u_{rms}^2$$

$$KE = \frac{1}{2} \rho u_{rms}^2$$

→ Relation b/w pressure and kinetic energy per unit

Volume =

$$\rightarrow \text{pressure} = \frac{1}{3} \rho u_{rms}^2 \dots (i)$$

$$\rightarrow \text{kinetic energy} = \frac{1}{2} \rho u_{rms}^2 \dots (ii)$$

(i) $\div$ (ii)

$$\frac{p}{KE} = \frac{\frac{1}{3} \rho u_{rms}^2}{\frac{1}{2} \rho u_{rms}^2}$$

$$\frac{p}{KE} = \left(\frac{\frac{1}{3}}{\frac{1}{2}} \right)$$

$$\frac{p}{KE} = \frac{2}{3}$$

$$p = \frac{2}{3} KE$$

→ Kinetic Interpretation of temperature =

Ideal gas equation for one mole

$$pV = RT \left[p = \frac{2}{3} KE \right]$$

$$\frac{2}{3} KE V = RT \quad \text{--- (1)}$$

Kinetic energy per unit volume $KE = \frac{1}{2} m u_{rms}^2$

from equation --- (1)

$$\frac{2}{3} \frac{1}{2} m u_{rms}^2 V = RT$$

$$\frac{1}{3} m u_{rms}^2 = RT$$

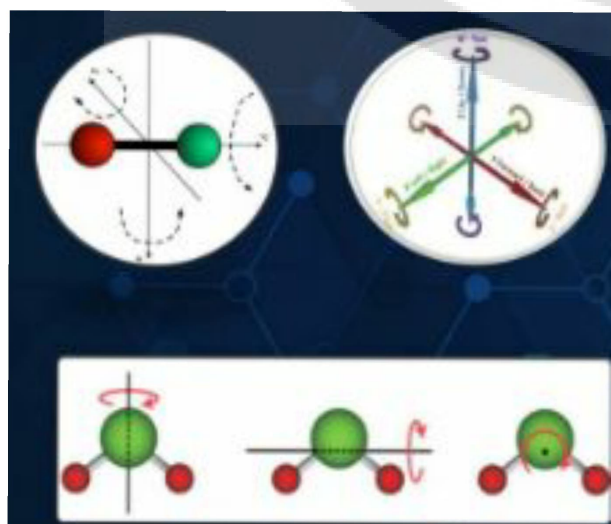
$$u_{rms}^2 = \frac{3RT}{m}$$

$$u_{rms} = \sqrt{\frac{3RT}{m}}$$

$$u_{rms} \propto \sqrt{T}$$

T = Temperature

→ Degrees of freedom and Law of Equipartition of Energy =



→ Degrees of freedom (f) = The number of independent ways in which a molecule can store energy.

→ There are three types of motion

- (i) Translatory
 - (ii) Rotatory
 - (iii) Vibrational
- } motion

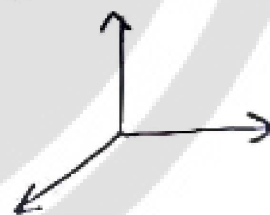
⇒ Types =

- ① mono atomic gases = He, Ne, Ar...
- ② Diatomic gas = H₂, O₂, N₂...
- ③ polyatomic gases = CO₂, NH₃

① Degree of freedom for mono atomic gases =

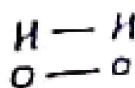
- (i) Translatory motion = 03
- (ii) rotational motion = 0

$$f = 3$$



② Degree of freedom for diatomic gases =

- (i) Translatory motion = 03
- (ii) Rotation motion = 02



$$f = 03 + 02 = 05$$

→ Degree of freedom for polyatomic =

(i) Linear = CO₂



Same as diatomic

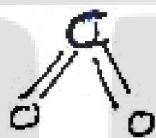
$$f = 5$$

(ii) Non linear = CO₂

(a) Translatory motion = 03

(b) Rotation motion = 03

$$f = 6$$



→ Degrees of freedom and Law of Equipartition of Energy =

→ It is given by Maxwell
→ Energy is directly proportional to Temperature.

→ Law of Equipartition of Energy = Energy is equally distributed among all degrees of freedom

→ The total energy of gas molecules is

$$U dT$$

$$U = \frac{1}{2} K_B T$$

→ Kinetic energy of one molecule with one degree of freedom

$$= \frac{1}{2} K_B T$$

→ Kinetic energy for 'f' degree of freedom for one molecule

$$KE = \frac{f}{2} K_B T$$

K.E. for N molecule or one mole

$$= N K.E.$$

$$= N \frac{f}{2} K_B T$$

$$\text{we know } K_B = \frac{R}{N}$$

$$R = K_B N$$

So

$$K.E. = \frac{f}{2} RT$$

Specific Heat at Constant Volume = $C_V = \frac{dE}{dT}$

E = Energy
T = Temperature

Specific Heat for monoatomic gas

$$E = \frac{f}{2} RT \quad f = 3$$

$$E = \frac{3}{2} RT$$

$$\Rightarrow C_V = \frac{dE}{dT}$$

$$C_V = \frac{d}{dT} \left(\frac{3}{2} RT \right)$$

$$C_V = \frac{3}{2} R \frac{dT}{dT}$$

$$C_V = \frac{3R}{2}$$

from Mayer's Relation =

$$C_P - C_V = R$$

$$\left[C_V = \frac{3R}{2} \right]$$

$$C_p - \frac{3}{2}R = R \quad \left| \quad \gamma = \frac{C_p}{C_v} \right.$$

$$C_p = R + \frac{3}{2}R \quad \left| \quad \gamma = \frac{\frac{5}{2}R}{\frac{3}{2}R} \right.$$

$$\boxed{C_p = \frac{5}{2}R} \quad \left| \quad \gamma = \frac{5}{3} = 1.66 \right.$$

Specific Heat for diatomic gas

$$E = \frac{f}{2} RT \quad (f=5)$$

$$E = \frac{5}{2} RT$$

$$C_v = \frac{dE}{dT}$$

$$C_v = \frac{d}{dT} \left(\frac{5}{2} RT \right)$$

$$C_v = \frac{5}{2} R \frac{dT}{dT}$$

$$C_v = \frac{5}{2} R$$

→ Mayer's relation

$$C_p - C_v = R \quad \left| \quad \gamma = \frac{C_p}{C_v} \right.$$

$$C_p - \frac{5}{2}R = R \quad \left| \quad \gamma = \frac{\frac{7}{2}R}{\frac{5}{2}R} \right.$$

$$C_p = R + \frac{5}{2}R$$

$$\boxed{C_p = \frac{7}{2}R}$$

$$\gamma = \frac{7}{5} = 1.4$$

Specific Heat for triatomic gas

$$E = \frac{f}{2} RT$$

$$f=6$$

$$E = \frac{6}{2} RT$$

$$E = 3RT$$

$$C_v = \frac{dE}{dT}$$

$$C_v = \frac{d}{dT} 3RT$$

$$C_v = 3R \frac{dT}{dT}$$

$$C_v = 3R$$

→ Mayer's Relation

$$C_p - C_v = R$$

$$C_p - 3R = R$$

$$C_p = 3R + R$$

$$C_p = 4R$$

$$\gamma = \frac{C_p}{C_v}$$

$$\gamma = \frac{4R}{3R}$$

$$\boxed{\gamma = 1.33}$$

⇒ Application to Specific Heat Capacities of gases

→ The molar specific heat capacity at constant volume (C_v) is given by.

→ The molar specific heat capacity at constant pressure (C_p)

→ The ratio $\gamma = \frac{C_p}{C_v}$ is called the adiabatic index

⇒ Degrees of freedom for different gases

Type of Gas	Degrees of Freedom (f)	C_v	C_p	γ
Monoatomic (He, Ne)	3 (Translational)	$(3/2)R$	$(5/2)R$	5/3
Diatomic (O_2 , N_2)	5 (3 Translational + 2 Rotational)	$(5/2)R$	$(7/2)R$	7/5
Polyatomic (CO_2 , CH_4)	6 (3 Translational + 3 Rotational)	3R	4R	4/3

⇒ Mean free path = The distance covered by gas molecule b/w two successive collision, is called path (λ)

$$\text{mean free path} = \frac{\text{Distance covered in time 't'}}{\text{no. of collision in 't' time.}}$$

Let us assume n no. of molecules are present in cylindrical shape of Area A and length L , and assume that all molecules moving with v velocity and collide with other molecules



mean free = distance covered in time t
no. of collision

$$d = \frac{v \times t}{n \times \text{volume}}$$

$$d = \frac{v \times t}{n \times \pi d^2 \times L} \quad \text{--- (1)}$$

[volume = Area $\times$ length]

$$[v = \pi d^2 \times L]$$

$$(L = v \times t)$$

$$d = \frac{v \times t}{n \times \pi d^2 \times v \times t}$$

$$d = \frac{1}{n \pi d^2}$$

When all molecules are at Rest

$\Rightarrow$ Maxwell's gave formula when all molecules are in motion

$$d = \frac{1}{\sqrt{2} n \pi d^2}$$

$\Rightarrow$ Avogadro's number (N_A) =

$\rightarrow$ Avogadro's number is the total number of molecules in one mole of any substance

$\rightarrow$ Relation between Avogadro's number and Boltzmann's Constant $[K_B = \frac{R}{N_A}]$

$\rightarrow$ where $K_B = 1.38 \times 10^{-23} \text{ J/K}$ is Boltzmann's constant.

$\Rightarrow$ Conclusion = The Kinetic theory of gases provides a microscopic explanation of pressure temperature and gas behavior.

$\rightarrow$ The Equipartition theorem helps determine specific heat capacities.

$\rightarrow$ The rms speed and mean free path describe the motion of gas molecules.

$\rightarrow$ Avogadro's number connects the molecular and macroscopic properties of gases.

$\rightarrow$ These Concise notes cover all key concepts in a structured and exam-oriented

manner let me know
If you need any modification.

Example = The temperature of a gas is 127°C . The volume of this gas is increased 1.5 times at constant pressure. How much would be the temperature of the gas?

$$T_1 = 127^{\circ}\text{C}$$

$$T_1 = 127 + 273$$

$$T_1 = 400\text{K}$$

$$p = \text{constant}, T_2 = ?$$

$$V_2 = 1.5 V_1$$

$$\frac{V_2}{V_1} = \frac{1.5}{1}$$

Charles's law

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$T_2 = \frac{V_2}{V_1} T_1$$

$$T_2 = \frac{1.5}{1} \times 400$$

$$T_2 = 600\text{K}$$